Zhanhong He

Audio Generative Modelling | Music Model Post Training | Perceptual Evaluation

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Profile

Music and audio machine learning researcher with doctoral research in musical expression and research internships at Huawei and Dolby. Experience spans DPO/GRPO-based music-model post-training, music-specific rewards and preference data, generative modelling, and human listening evaluation. Works with audio and symbolic music models in Python and PyTorch.

Research And Industry Experience

Research Intern - Huawei Hongkong Research Centre | Started June 2026

- Applied DPO and GRPO to music-generation post-training, including rollout workflows and music-specific reward design.
- Developed rewards informed by music information retrieval and signal processing, and contributed to human-preference data collection and analysis.

Research Intern - Dolby Laboratories, Sydney | May-September 2024

- Developed music-generation models using U-Net, variational autoencoder and masked Transformer architectures.
- Designed and conducted professional human listening tests to evaluate generated musical expression.

Doctoral Research - University of Western Australia | February 2022-present

- Contributed to research on multi-task CRNN models for musical dynamics, beats, downbeats and structural changes, including a PyTorch implementation of a Bark-scale specific-loudness feature extractor.
- Contributed to cross-instrument MIDI-velocity estimation research using differentiable SoundFont proxies and audio-based supervision in settings with limited velocity labels.
- Contributed to a score-informed Transformer approach for refining MIDI-velocity estimates from automatic music transcription systems.
- Contributed to adapting a U-Net image-colourisation approach to MIDI-velocity prediction and to a listening evaluation with professional listeners.

Tutor and Lab Facilitator - University of Western Australia | February 2023-December 2025

- Taught Signal Processing and Engineering Practices, and co-supervised a Computer Science Honours project.

Selected Publications And Research

He, Z., Meng, H., Huang, D., and Togneri, R. Joint Estimation of Piano Dynamics and Metrical Structure with a Multi-Task Multi-Scale Network. ICASSP, 2026. DOI: 10.1109/ICASSP55912.2026.11460839.

He, Z., Meng, H., Huang, D. D., and Togneri, R. Beyond Piano: Cross-Instrument MIDI Velocity Estimation via Differentiable SoundFont Proxies. arXiv:2608.08985, 2026. https://arxiv.org/abs/2608.08985

Score-Informed Transformer for Refining MIDI Velocity in Automatic Music Transcription. Research project and implementation: https://github.com/zhanh-he/score-informed-amt

Filling MIDI Velocity using U-Net Image Colorizer. Research project and implementation: https://github.com/zhanh-he/midi-velocity-colorizer

Education

PhD research in Electrical and Electronic Engineering - University of Western Australia | February 2022-present

Doctoral defence completed; graduation certificate expected November 2026.

BSc (Honours) in Engineering Science - University of Western Australia | October 2018-December 2021

First Class Honours | GPA 7/7 | WAM 87.7/100

Honours thesis: Improving Spectrogram-based Music Source Separation using Phase Information.

Technical Practice

Python; PyTorch; audio and symbolic music processing; DPO/GRPO; music reward design; human-preference data; listening evaluation; U-Net, VAE, Transformer and CRNN research implementations.

Ableton Live; Max/MSP; MuseScore. Guitar and piano as a music enthusiast.